Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	4 April 2026
Team ID	SWTID-2026-3550
Project Name	Streamlining Ticket Assignment For Efficient Support Operations
Maximum Marks	4 Marks

Technical Architecture (Description)

The system follows a 3-tier architecture:

1. **Presentation Layer (UI):**
   Users (customers, support team, admin) interact through a web interface.
2. **Application Layer (Backend Logic):**
   Handles ticket creation, classification, automated routing, role-based access, and workflows.
3. **Data Layer (Database):**
   Stores user data, tickets, roles, groups, and workflow configurations.

Flow:

User → Web UI → Backend Application → Database

→ Automated Routing Engine → Assigned Support Group → Status Updates → User

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for users, support team & admin	HTML, CSS, JavaScript, React JS
2	Application Logic-1	Ticket creation & management	Python / Node.js
3	Application Logic-2	Automated ticket routing & workflow engine	ServiceNow Flow Designer / Backend Logic
4	Application Logic-3	Role-based access control & authentication	ServiceNow Roles / JWT Authentication
5	Database	Stores users, tickets, roles, groups	MySQL / ServiceNow Tables
6	Cloud Database	Cloud-based data storage	ServiceNow Cloud
7	File Storage	Stores attachments in tickets	Cloud Storage / ServiceNow Attachments
8	External API-1	Email notification service	SMTP / SendGrid API
9	External API-2	Authentication or integration service	Google OAuth API

10	Machine Learning Model	(Optional) Ticket classification automation	Basic ML Model / Future Scope
11	Infrastructure (Server / Cloud)	Deployment environment	ServiceNow Platform / Cloud (AWS/Azure)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for frontend/backend	React JS, Node.js
2	Security Implementations	Secure login, role-based access, encryption	JWT, HTTPS, RBAC, ServiceNow ACL
3	Scalable Architecture	System can handle increasing users & tickets	3-Tier Architecture, Cloud Infrastructure
4	Availability	System available with minimal downtime	Cloud Hosting, Load Balancing
5	Performance	Fast ticket processing and response time	Caching, Optimized Queries

Summary

This architecture ensures:

- **Efficient automated ticket routing**
- **Strong security and access control**
- **High scalability and availability**
- **Reliable performance for real-time operations**